

Fig. 2: Interconnection of cigar plug and USB power socket using a coil cord

circuitry. Usually 5V DC, 100mA current is required to satisfy the input power demand.

Fig. 1 shows the circuit of a versatile USB power socket that safely converts the 12V battery voltage into stable 5V. This circuit makes it pos-

sible to power/recharge any USB power-operated device, using in-dash board cigar lighter socket of your car.

The DC supply available from the cigar lighter socket is fed to an adjustable, three-pin regulator LM317L (IC1). Capacitor C1 buffers any disorder in the input supply. Resistors R1 and R2 regulate the output of IC1 to steady 5V, which is available at the 'A' type female USB socket. Red LED1 indicates the output status and zener diode ZD1 acts as a protector against



Fig. 3: Pin configuration of LM317L (To-92 package)

high voltage.

Assemble the circuit on a general-purpose PCB and enclose in a slim plastic cabinet along with the indicator and USB socket. While wiring the USB outlet, ensure correct polarity of the supply.

For interconnection between the cigar plug pin and the device, use a long coil cord as shown in Fig. 2. Pin configuration of LM317L is shown in Fig. 3. ●

CAR FAN SPEED CONTROLLER

■ P.V. VINOD THEKKUMURI

Using this circuit you can control the speed of 12V DC fans used in cars. The circuit is built around timer 555, which is

wired as an astable multivibrator. The output of the multivibrator is fed to IRF 540 MOSFET. The fan is connected between the positive terminal of the battery and drain (D) of MOSFET T1. Capacitor C1 is connected in parallel

to the fan to stabilise its speed. Free-wheeling diode D1 protects the motor from back emf. A

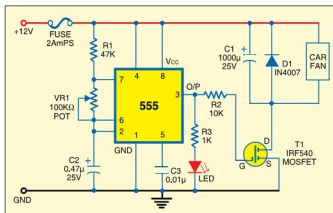


Fig. 1: Car fan speed controller



Fig. 2: Pin configuration of IRF540

fuse is included for protection.

Potmeter VR1 is used to change the duty cycle of the multivibrator and hence the speed of the fan. If you feel that low/high level of the fan speed is not sufficient, increase/decrease the value of C2 (0.47 µF) to reduce/increase the speed of the fan.

Assemble the circuit on a general-purpose PCB and enclose in a suitable cabinet. Fix the potmeter at the front side of the case, so that you can easily change the speed of the fan. Connect the battery to the fan using wires with suitable current-carrying capacities. Use red wire for positive terminal and black wire for negative terminal. ●

DIGITALLY ADJUSTABLE DANCING LIGHTS

■ ABHISHEK KUMAR SINGH

You might have come across several types of adjustable dancing lights (flickering LEDs). Most of them use presets (variable resistors) to adjust the rate of switching. Being a mechanical component, the preset easily

wears out with use and also introduces noise in the circuit. The circuit presented here selects different values of resistors to control the frequency of an astable multivibrator using timer IC 555.

The circuit is built around decade counter IC 4017, quad bilateral switch IC 4066 (to select the desired resist-

ance) and timer 555. The decade counter output selects one of the resistors